

Ultrafiltration

Final Lab Report
Unit Operation Lab 4
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Group 1



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Executive Summary

The objective of this experiment is to measure the flux through the membrane for different pressure drops and for different milk concentration [1]. The amount of time to collect the permeate increased with increasing percentage recovery and decreasing pressure drop. The water flow rate was the greatest with pure water and decreased with increasing percentage recovery. As the percentage recovery increased, the total volumetric flow rate of the permeate flowing through the membrane decreased. As the pressure drop increased, the total volumetric flow rate of permeate increased as well. As the percentage of recovery and pressure drop was increased, the total volumetric flow rate of permeate decreased. The concentration factor increased as the percentage recovery was increased as well. The average apparent permeability decreased as the percentage of recovery was increased. As the percentage recovery of the permeate was increased, the volumetric flux across the membrane would both increase and decrease not giving a direct correlation. The amount of time to collect the permeate increased with time because with increased recovery comes a more concentrated permeate to collect. From figure 2, as pressure drop was increased, the volumetric permeate flow rate decreased. This was consistent with the prediction that the net driving pressure differential across the membrane is proportional to the permeate flow rate. In figure 2 of the permeate flux vs. average applied pressure within the column, the permeate flux decreased with increasing percentage recovery of permeate. This was not consistent with theory and may have resulted from membrane fouling. Membrane fouling is predicted to start with pore blocking that is followed by a constant cake formation located on the surface of the membrane. Also in figure 2, the permeate flux decreased with increasing pressure drop. This was not expected because the pressure difference across the membrane is what drives the water flow through the membrane [6]. This inconsistent correlation may have resulted from the fact that flux rate will also increase when pressure is increased making the system more prone to fouling and require more chemical cleaning [7].

Introduction

The purpose of this laboratory experiment is to analyze an ultrafiltration system and see how effective the filtering is [1]. Ultrafiltration is a relatively new process in the field of chemical engineering. Ultrafiltration can be used to concentrate solutions that contain small macromolecules or colloidal sized particles [1]. It emerged in the 1970s as polymer chemists and engineers discovered new methods to fabricate porous polymeric membranes that can accomplish separation on a molecular scale. Currently, it is used in the purification of natural products, recovery for pharmaceuticals, and applications for environmental management [1]. The experimental apparatus includes a hollow fiber module connected to tanks for the process feed and the permeate product. Next, the feed pump delivers a constant head of pressure and valves on the inlet and outlet to the module where they are adjusted to fix the inlet and outlet pressure [1]. Before beginning the actual experiment the filter has to be cleaned. Then a water flux test to ensure that the filter is clean. After the cleaning cycle is completed, the experiment can begin. Using a milk-water mixture, ultrafiltration is used remove the solvent from solution to concentrate the milk (solute). Then volumetric flow rates for the ultrafiltration process are to measured at several pressure differentials. Different milk concentrations are obtained through the following ways. The feed is first run through the membrane until the appropriate amount of solvent (permeate) is removed. From this point, the permeate is directed back to the feed tank and the flux measurement is then made. After the permeate flow is directed back to the tank, no

further recovery is obtained and the feed concentration will not change with time. This will allow for the measurement of the permeate flux at constant feed concentration. Ultrafiltration is considered to be one of the most impactful processes in the separation field today. It is widely used in major industries like the food, beverage, biotechnology, and pharmaceutical industries [2]. For example, in the dairy industry, during the production of cheese whey would be discharged to the sewer because of the high salt and lactose content making direct use as a food supplement more difficult [3]. With ultrafiltration, whey can now be processed to obtain additional food values by increasing the fraction of milk proteins used as the cheese and be able to reduce the amount of waste disposal from the whey [3].

Theory

The following mass balance can be applied to the ultrafiltration system:

$$\text{Mass feed} = \text{Mass concentrate} + \text{Mass permeate}$$

Where:

- Mass concentrate = the macromolecular solutes, colloids, dispersed phase, or any other species that is too large to pass through the membrane surface
- Mass permeate = the fluid solution that passes through the membrane surface.

For this experiment, the volumetric flowrates for an ultrafiltration process are measured at many pressure differentials. Many runs are made at different pressure differentials in which a percentage of the solution is removed. Finally, a graph of the Concentration Factor (CF) versus the flux is plotted in order to determine the optimum operating pressure for the filter.

Concentration Factor (CF) defined as:

$$CF = \frac{V_{\text{original}}}{V_{\text{concentrate}}} = \frac{V_{\text{original}}}{V_{\text{original}} - V_{\text{permeate}}}$$

Concentration factor can also be defined as

$$CF = \frac{1}{1 - R}$$

Where:

- R = recovery where the feed stream becomes the permeate

$$R = \frac{V_{\text{permeate}}}{V_{\text{original}}}$$

Permeate flux will be measured for various recoveries and different pressures. Flux is the rate of permeate that flow through a unit area of the membrane surface. A plot of the permeate flux versus concentration factor is used to determine the pressure differential where the greatest flux is obtained.

The basic model of ultrafiltration expresses the solvent flux through the membrane as a function of the applied pressure:

$$v = Q_v \Delta P$$

Where:

- Q_v = The membrane permeability for the solvent
- v = volume flux

This expression assumes that the solution is sufficiently dilute that osmotic pressure effects do not influence the driving force. It also assumes that the solutes do not permeate through the membrane. However, the non-permeating solute does concentrate at the membrane surface which is also known as concentration polarization.

At steady state, the convective flow of solute towards the surface must equal its back solution into the bulk. This mass balance over a thin element near the surface will give:

$$C_1 v = -D_v \frac{dC_1}{dx}$$

Where:

- D_v = The solute diffusion coefficient
- C_1 = The solute concentration in the bulk of the solution

Subject to the two boundary conditions, the solution of the differential equation gives the flux as

$$v = k_c \ln\left(\frac{C_s}{C_1}\right)$$

Where

- C_s = Solute concentration at the surface
- k_c = mass transfer coefficient ($\frac{D_v}{\delta}$)
- δ = thickness of concentration boundary layer
- D_v = Diffusion coefficient

Flux does not depend on pressure, it depends upon a balance of convection and diffusion in the boundary layer. This equation will show that the solvent flux versus the logarithm of the feed concentration should be a straight line if the surface concentration is constant. This equation also provides a simple test for when the concentration polarization is important in the ultrafiltration process. If C_s is small, then:

$$\frac{k_c}{v} \gg 1$$

In many mass transfer processes, the boundary layer can be estimated to be in the range of 0.01cm.

For % Recovery:

$$\% \text{ Recovery} = \frac{\text{Volume Permeate}}{\text{Volume Original}} * 100\%$$

Results

For the first part of the experiment, approximately 1000 mL of permeate was collected for 0%, 12%, 25%, and 40% recovery. As the amount of recovery percentage increased, the amount of time to collect the permeate increased as well with decreasing pressure drop. The water flow rate for pure water was the greatest at 2.708 L/min. The total volumetric flow rate of permeate flowing through the membrane decreased with increasing percentage recovery. The total volumetric flow rate of permeate increased with increasing pressure drop. The concentration factor increased with increasing percentage recovery of the permeate as well. From the pure water and milk/water solution, the average apparent permeability was calculated for each percentage recovery of permeate. As the recovery percentage increased the average apparent permeability decreased. The volumetric flux across the membrane decreased with increasing

percentage recovery of the permeate. The volumetric flux across the membrane would first increase then decrease not giving a clear correlation with the pressure drop.

Discussion

For the first part of the experiment that amount of time it took to collect the permeate increased with increasing percentage recovery. This was expected because with increased recovery comes a more concentrated permeate to collect. The water flow rate for pure water was the greatest because there was no concentrated permeate in the pure water. In figure 1, the volumetric permeate flow rate decreased with increasing pressure drop. This was consistent with the prediction that the permeate flow rate is proportional to the net driving pressure differential across the membrane. In figure 2 of the permeate flux vs. average applied pressure within the column, the permeate flux decreased with increasing percentage recovery of permeate. This was expected because of membrane fouling [4]. This is due to the fact that water is assumed to pass by laminar flow through the small pores of the selective layer, and the driving force is the pressure difference minus the difference in osmotic pressures across the membrane. Membrane fouling is believed to start with pore blocking followed a constant cake formation located on the surface of the membrane [5]. In figure 2, the permeate flux decreased with increasing pressure drop. This was not consistent with the experimental prediction because the pressure difference across the membrane is what drives the water flow through the membrane [6]. This inconsistent correlation may have been due to the fact the flux rate will also increase when pressure is increased making the system more prone to fouling and require more chemical cleaning [7]. To conclude, the amount of time to collect the permeate increased with time because with increased recovery comes a more concentrated permeate to collect. Therefore, obtained shows how ultrafiltration depends upon the relative rate of permeation of components through the membrane.

References

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- [5] Bahnasawy, Adel. "THEORETICAL AND EXPERIMENTAL STUDIES ON ULTRAFILTRATION (UF) PROCESS FOR MILK CONCENTRATION." *Academia.edu*, www.academia.edu/8320981/THEORETICAL_AND_EXPERIMENTAL_STUDIES_ON_ULTRAFILTRATION_UF_PROCESS_FOR_MILK_CONCENTRATION.
- [6] *Membrane Systems Educational Module*, web.utk.edu/~qhe2/MembraneModule/Theory.html.

[7] Nemeth-Harn, Julie. "How to Calculate Percent Recovery and What Affects It in a Membrane Treatment System?" *Harn*, blog.harnrosystems.com/how-to-calculate-percent-recovery-and-what-affects-it-in-a-membrane-treatment-system.

Appendix I: Data Tabulations/graphs

	Seconds per 1000 mL				
Outlet Pressure [psig]	5	15	10	20	5
Pressure Drop [psig]	20	10	15	5	20
0% recovery	22.16	16.61	18.15	14.76	22.03
12% recovery	33.61	25.52	29.38	23.15	36.92
25% recovery	35.32	25.98	31.59	24.92	38.04
40% recovery	38.48	27.36	32.42	25.07	40.14

Table 1: Time taken for 1000 mL of permeate to be collected

		Pressure Drop					Concentration factor
	Outlet Pressure	5	15	10	20	5	
	Pressure Drop [psig]	20	10	15	5	20	(CF)
Pure water	Water Flowrate (L/min)	2.708	3.612	3.305	4.064	2.724	1.000
12% recovery	Water Flowrate (L/min)	1.785	2.351	2.042	2.592	1.625	1.136
25% recovery	Water Flowrate (L/min)	1.699	2.310	1.899	2.407	1.577	1.333
40% recovery	Water Flowrate (L/min)	1.559	2.193	1.851	2.394	1.495	1.667

Table 2: Total volumetric flowrate of permeate flowing through the membrane. Included is the concentration factor for each recovery.

%Recovery	Average Apparent Permeability
0	2.41E-05
12	1.52E-05
25	1.45E-05
40	1.38E-05

Table 3 : Apparent permeability under conditions of pure water and milk/water solution

	Pressure Drop [psig]	20	10	15	5	20	(CF)
Pure water	Water Flux (L/min*m m ²)	0.000356	0.000475	0.000435	0.000535	0.000358	1.000
12% recovery	Water Flux (L/min*m m ²)	0.000235	0.000309	0.000269	0.000341	0.000214	1.136
25% recovery	Water Flux (L/min*m m ²)	0.000224	0.000304	0.000250	0.000317	0.000208	1.333
40% recovery	Water Flux (L/min*m m ²)	0.000205	0.000289	0.000244	0.000315	0.000197	1.667

Table 4: Volumetric flux across the membrane. Included is the concentration factor for each recovery.

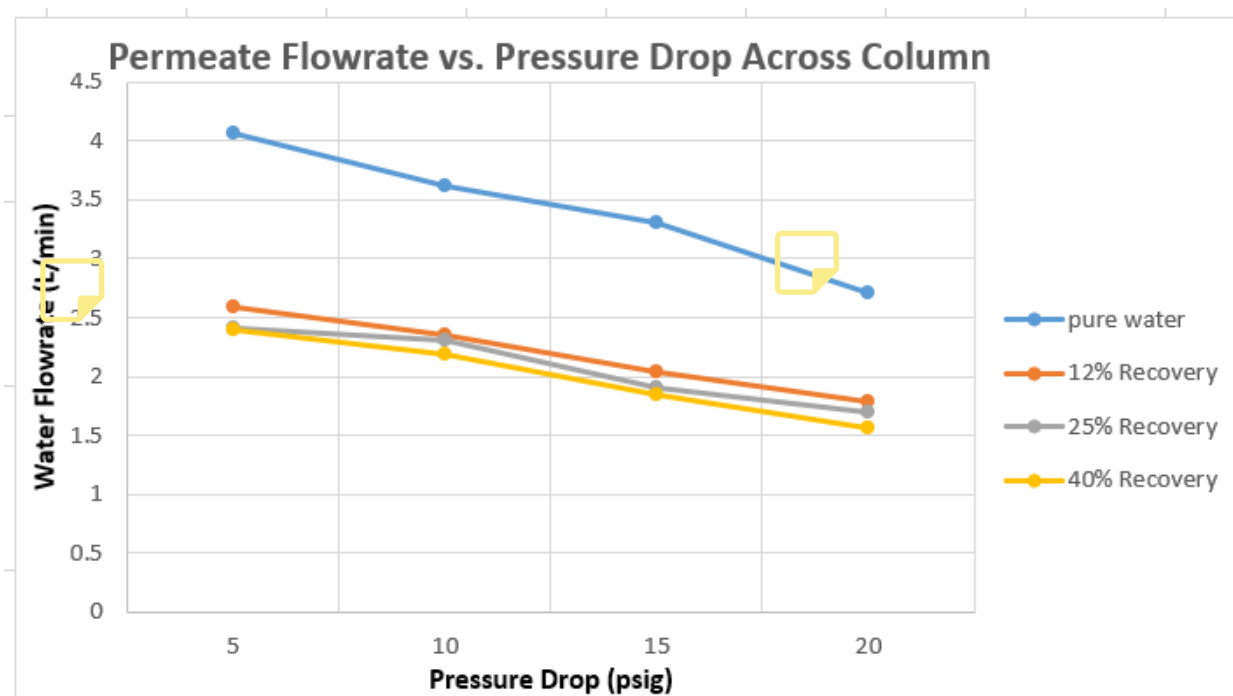


Figure 1 : volumetric flow rate through the filtration membrane as a function of pressure drop. Four different recoveries were analyzed.

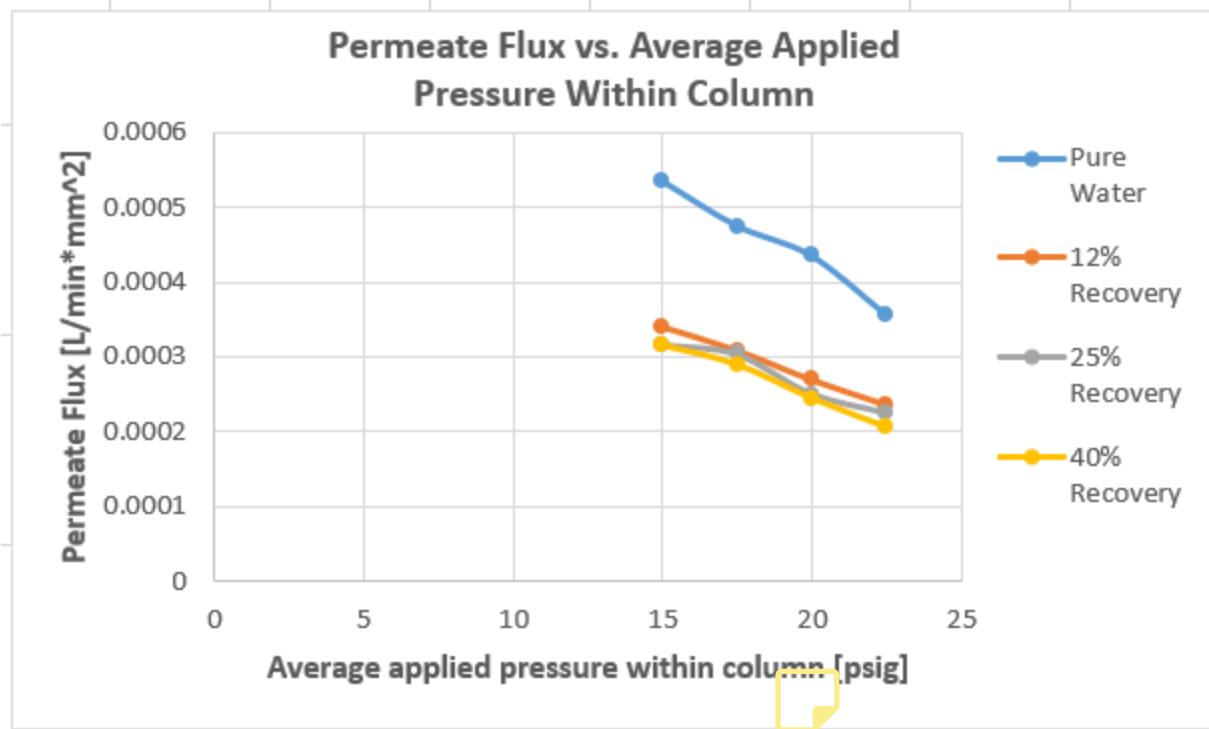


Figure 2: Permeate Flux vs. Average Applied Pressure Within the capillary tubes for skim milk solutions and the pure water

Appendix II: Error Analysis

- The pressure vs flux graph has some discrepancy, but the general trend can be noticed. The membrane had to be rigorously cleaned through multiple cycles, but since milk is used in couple trials, there is a possibility of bacterial growth within the membrane. It is suspected that the membrane wasn't cleaned properly, which is why the results have had some discrepancy. It is suggested to incorporate another set of rigorous cleaning or installing a new membrane for better generation of results.
- In order to get the flux, one had to collect the volume of ~1000 mL and the time it took to fill 1000 mL by the aid of a stopwatch. While acquiring this data, the volume was read off by the naked eye and there were some observational errors that have caused some discrepancy in the generation of the graphs of pressure vs flux. These errors are inevitable as they are human errors.

- Another source of observational error would have occurred while setting the flow valves to get the exact pressures at the top and bottom, which would affect the flux considerably. This causes discrepancy in the generation of the pressure vs flux graph.
- There are no readings on the process tank to exactly acquire the volume; the readings were hand drawn. This will then translate into more errors when creating the 2.0 pH phosphoric acid solution, the sodium hydroxide (1 wt%), and the 200 ppm sodium hypochlorite solution in the cleaning processes. Even though none of these solutions are used for experimental data they would have an effect on the skim milk trials later on if these cleanings did not properly flush the system. Considering Bio-fouling in the membranes even though so much of each solution was flushed through the system, it still affects the cleaning. This is a random error and can be corrected by installing a reading scale on the process tank and setting up an electric field with in the membrane.
- While collecting data for the flux, another error occurred as collected fluid was transferred from the hose to the graduated cylinder and the process tank, some of the fluid was lost due to human error. It was also noted that there were small bubbles entrapped within the membrane, which could have affected the actual flux of the system. This is a human error and can be corrected by careful handling during the process of acquisition of data for the flux.
- Last but not least, after analyzing the results the main source of error would have been human error and bacterial growth within the membrane as it affects the data acquired for the flux. If replacing the membrane isn't feasible, setting up an electric field with in the membrane could be good alternative.

Appendix III: Sample Calculations**Water Flux:**

$$\text{Flow rate} = \frac{1000\text{mL}}{\text{seconds}} * \frac{60\text{sec}}{\text{min}} * \frac{1\text{L}}{1000\text{mL}}$$

$$\frac{1000}{22.16} * \frac{60}{1} * \frac{1}{1000} = 2.708 \text{ L/min}$$

Concentration Factor

Total Volume of Milk & Water (L) = 83.785

$$CF = \frac{\text{total volume}}{\text{total volume} - (\% \text{ recovery} * \text{total volume})}$$

$$CF = \frac{83.78}{83.78 - (0.12 * 83.78)} = 1.136$$

CF=Concentration Factor (for the 12% Recovery)

Permeability

Volumetric Flux = Permeability * Pressure Drop

$$v = Qv * \Delta P$$

$$Qv = 0.000356 / 15 = 0.00002375 \text{ mm}^2$$

Appendix IV: Individual Team Contributions**Raghad Albadri**

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Completed
Pre-Lab Editing	1	Completed
Experimental Objective (Prelab)		
Apparatus (prelab)	1	Completed
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
Conclusion (final lab)		

References (final and/or prelab)		
Data Tabulation/Graphs (final)	5	Completed
Error Analysis (final lab)	1	Completed
Sample Calculations (final Lab)	0.5	Completed
PowerPoint Presentation		
Total	16.5	

Ekaterina Ivanov

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Completed
Pre-Lab Editing	1	Completed
Experimental Objective (Prelab)	1	Completed
Apparatus (prelab)	1	Completed
Materials & Supplies (prelab)	1	Completed
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1	Completed

Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)	1	Completed
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation		
Total	14	Completed

Irene Jose

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Completed
Pre-Lab Editing	1	Completed

Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)	1	Completed
Final Lab Editing		
Executive Summary (final lab)	1	Completed
Introduction (Final lab)	1	Completed
Theory (Final Lab)	1	Completed
Results (final lab)	1	Completed
Discussion (final lab)	1	Completed
Conclusion (final lab)	1	Completed
References (final and/or prelab)	1	Completed
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		

Power Point Presentation		
Total	17	Completed

David Schulz

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Completed
Pre-Lab Editing	1	Completed
Experimental Objective (Prelab)		
Apparatus (prelab)	1	Completed
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	3	Completed
Discussion (final lab)		

Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	1	Completed
Error Analysis (final lab)	1	Completed
Sample Calculations (final Lab)		
Power Point Presentation		
Total	15	Completed